Fitting a Weibull to the period of $\sqrt{n}$

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1 The problem

Every square root of a non square whole number repeats forever.

$\sqrt{2}$	$[1;\ \overline{2}]$
$\sqrt{3}$	$[1;\ \overline{1,2}]$
$\sqrt{5}$	$[2;\ \overline{4}]$
$\sqrt{6}$	$[2;\ \overline{2,4}]$
$\sqrt{7}$	$[2;\ \overline{1,1,1,4}]$
$\sqrt{13}$	$[3;\ \overline{1,1,1,1,6}]$
$\sqrt{19}$	$[4;\ \overline{2,1,3,1,2,8}]$
$\sqrt{31}$	$[5;\ \overline{1,1,3,5,3,1,1,10}]$
$\sqrt{94}$	$[9;\ \overline{1,2,3,1,1,5,1,8,1,5,1,1,3,2,1,18}]$

The notation $[a; b, c, d, \ldots]$ means the tower

$$a + \cfrac{1}{b + \cfrac{1}{c + \cfrac{1}{d + \cdots}}}$$

and the bar marks the part that repeats. Write $L(n)$ for the length of that repeating block.

The length is not predictable from n. $\sqrt{61}$ needs eleven terms and $\sqrt{65}$ needs one. No formula gives it. So the question is statistical rather than exact: taken over many n, what distribution does the period follow?

$L(n)$ itself grows, roughly like $\sqrt{n}$, so the variable to study is the normalised one

$$t \;=\; \frac{L(n)}{\sqrt{n}}.$$

Its density is a decaying curve with a soft right hand edge near 2.5. It looks exponential. This document asks whether it is, and fits the natural one-parameter generalisation.

2 The Weibull distribution

Write $S(t) = P(X > t)$ for the survival function. The exponential and the Weibull sit side by side:

$$\text{exponential} \quad S(t) = e^{-t/\lambda}, \qquad\qquad \text{Weibull} \quad S(t) = e^{-(t/\lambda)^k}.$$

One extra exponent. Its meaning appears in the hazard rate $h(t) = f(t)/S(t)$, the instantaneous rate of stopping given that you have got this far:

$$\text{exponential} \quad h(t) = \frac{1}{\lambda}, \qquad\qquad \text{Weibull} \quad h(t) = \frac{k}{\lambda}\Big(\frac{t}{\lambda}\Big)^{k-1}.$$

So the Weibull is the distribution whose hazard is a power of t, and k is that power.

$k = 1$	constant hazard. The Weibull *is* the exponential. Memoryless.
$k < 1$	falling hazard. Survive a while and you are more likely to continue.
$k > 1$	rising hazard. Wear-out; a soft edge ahead.

It is the right second thing to try after an exponential because it adds exactly one parameter, and that parameter indexes the only feature the exponential refuses to have.

Note for later that h is a power of t, hence *monotone for every* k. A Weibull hazard can fall or rise. It cannot do one and then the other.

3 Method

The period. Generated by the PQa recurrence, which holds the surd in exact integers and never touches a float. With $a_0 = \lfloor\sqrt{n}\rfloor$, $P_0 = 0$, $Q_0 = 1$,

$$a_i = \left\lfloor \frac{a_0 + P_i}{Q_i} \right\rfloor, \qquad P_{i+1} = a_i Q_i - P_i, \qquad Q_{i+1} = \frac{n - P_{i+1}^2}{Q_i},$$

and the block closes at the first $i \geq 1$ with $Q_i = 1$. Every quantity is an integer, so there is no tolerance to choose and no accumulated error.

The fits. Maximum likelihood, with the location parameter fixed at 0 since $t > 0$ by construction. The exponential fit is then just the sample mean.

The test. Kolmogorov–Smirnov distance between the empirical and fitted distribution functions, compared against the 5% critical value $1.358/\sqrt{N}$.

The bands. Seven disjoint ranges of n from 10^4 to 9×10^5, each fitted separately, so that any drift in the fitted parameters is visible rather than averaged away.

4 Results

band	N	mean t	k	KS exp	KS Weib	5% cut
$[10^4, 4{\cdot}10^4)$	29,900	0.3872	1.0369	0.0222	0.0214	0.0079
$[5{\cdot}10^4, 8{\cdot}10^4)$	29,941	0.3632	0.9961	0.0248	0.0239	0.0078
$[10^5, 1.3{\cdot}10^5)$	29,956	0.3526	0.9826	0.0232	0.0191	0.0078
$[2{\cdot}10^5, 2.3{\cdot}10^5)$	29,968	0.3420	0.9596	0.0338	0.0216	0.0078
$[3{\cdot}10^5, 4{\cdot}10^5)$	99,915	0.3343	0.9480	0.0346	0.0210	0.0043
$[5{\cdot}10^5, 5.3{\cdot}10^5)$	29,979	0.3295	0.9393	0.0412	0.0217	0.0078
$[9{\cdot}10^5, 9.3{\cdot}10^5)$	29,984	0.3207	0.9264	0.0442	0.0193	0.0078

Three things are in that table.

The Weibull beats the exponential everywhere, and increasingly. At $n \sim 10^4$ the two are nearly tied, 0.0214 against 0.0222. By $n \sim 9 \times 10^5$ the Weibull is better by more than a factor of two, 0.0193 against 0.0442.

Neither is accepted anywhere. Every KS distance in both columns exceeds its critical value, the Weibull by between 2.4 and 4.9 times. These are good descriptions of the data, and they are not the law.

The shape parameter drifts, monotonically, and does not settle. It starts above 1, passes through 1, and keeps falling. A straight line in $\log n$ fits the seven values well:

$$k \;\approx\; 1.3197 - 0.02897 \log n, \qquad \text{crossing } k = 1 \text{ at } n \approx 61{,}900.$$

The mean drifts the same way, $\text{mean}\, t \approx 0.5552 - 0.01724 \log n$.

> The crossing is not a curiosity. Below $n \approx 6 \times 10^4$ the fitted hazard is *rising*; above it, *falling*. Any statement of the form "$L(n)/\sqrt{n}$ is approximately exponential" is therefore true at one scale and at no other, and the sign of the error reverses as you pass through it.

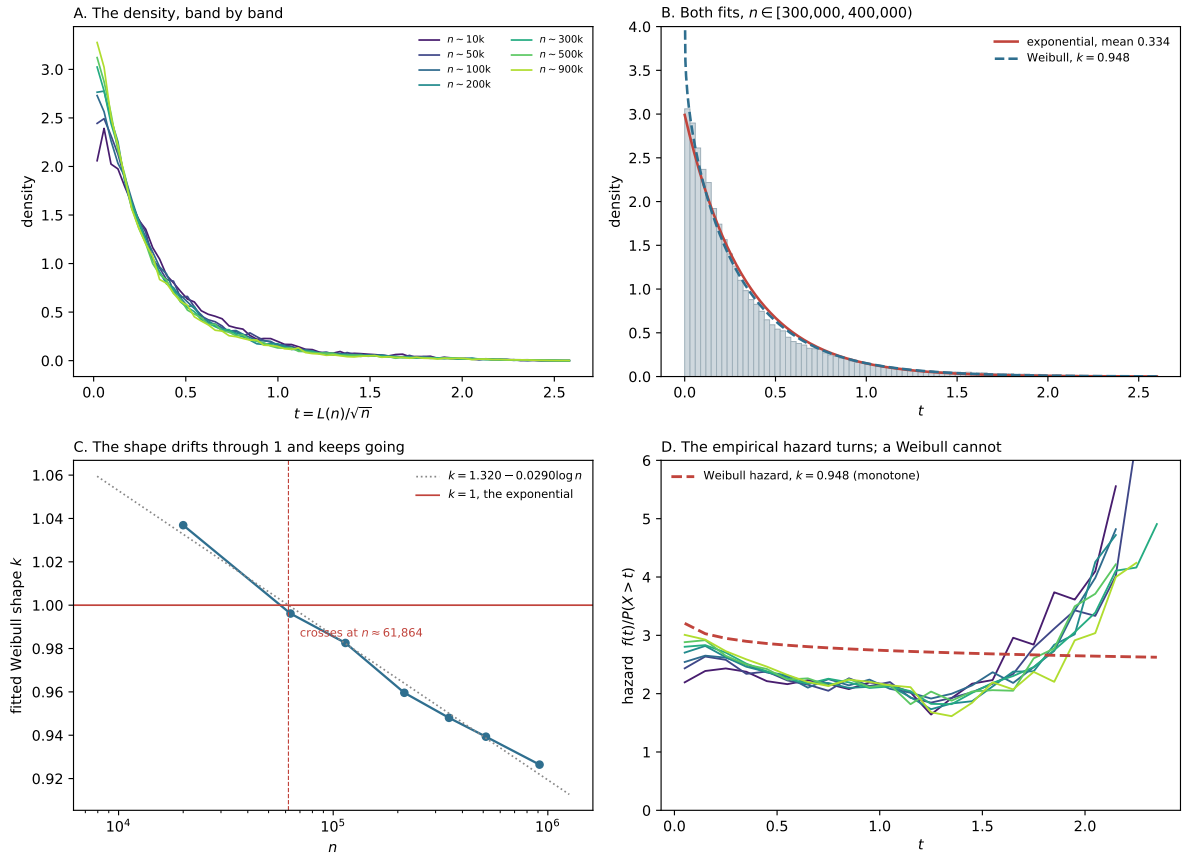
5 Why no Weibull fits

The fits are rejected, so it is worth seeing exactly what they miss. Take the empirical hazard on the largest sample, $n \in [3 \times 10^5, 4 \times 10^5)$, against the hazard of the Weibull fitted to it.

t	empirical hazard	Weibull, $k = 0.948$
0.00–0.10	2.804	3.206
0.60–0.70	2.169	2.806
1.20–1.30	1.825	2.712
1.40–1.50	2.014	2.691
1.90–2.00	3.056	2.650
2.30–2.40	4.907	2.625

The empirical hazard falls to a minimum near $t \approx 1.25$ and then climbs steeply, ending at 4.9 — it is U-shaped, and spans a factor of 2.7. The fitted Weibull runs from 3.21 down to 2.63 and nothing else, spanning a factor of 1.2, because a power of t is monotone.

So no member of the family can fit, whatever k is chosen. The estimator resolves the conflict by following the bulk, where nearly all of the likelihood is, and is therefore blind to the tail entirely. That is how a shape of 0.948 — which reads as "practically exponential" — still fails the test by a factor of five. The miss is not a slightly wrong curvature. It is a missing feature.



6 What the two arms are

The U has one mechanism per arm, and they are different.

The falling arm is a mixture. The period is essentially the regulator divided by the Lévy constant 1.187, and the spread across n comes from the class number. A mixture of exponentials with varying rates always has a falling hazard — the slow components survive longest, so survivors are drawn increasingly from them. Mixing cannot produce a rising hazard.

The rising arm must therefore be the edge. There is a ceiling, $L(n) = O(\sqrt{n}\log n)$; it is soft, and it drifts. As t approaches it the survivors run out of room. No mixture supplies that.

A single-parameter family is being asked to represent two mechanisms pulling in opposite directions, and the balance between them moves with n because the edge moves with n. That is why k drifts, and why the drift has no limit in this data.

7 How k depends on M, the largest n included

The fits above are on disjoint bands. A different and more natural question is what happens if you simply expand every square root up to some ceiling M and fit the lot — the cumulative sample $\{L(n)/\sqrt{n} : 2 \le n < M\}$.

Periods were computed for every non-square below 10^6 (25 seconds; 999,000 values, mean period 219.46) and the Weibull refitted at each of twenty-one values of M.

M	N	mean t	k	KS	5% cut
3,162	3,105	0.4770	1.2185	0.0293	0.0244
10,000	9,900	0.4365	1.1308	0.0237	0.0136
31,622	31,444	0.4052	1.0697	0.0214	0.0077
100,000	99,683	0.3771	1.0183	0.0207	0.0043
316,227	315,664	0.3539	0.9800	0.0199	0.0024
1,000,000	999,000	0.3348	0.9489	0.0190	0.0014

$$k(M) \approx 1.5517 - 0.04527 \log M, \qquad k = 1 \text{ at } M \approx 196{,}000.$$

It falls linearly in $\log M$ across the whole range, monotonically, with no sign of settling.

It is not the same line as the band-wise one. Against $k = 1.3197 - 0.02897 \log n$ from the previous section:

	band-wise	cumulative
slope in log	-0.02897	-0.04527
crosses $k = 1$ at	$n \approx 61{,}900$	$M \approx 196{,}000$

Two effects, pulling opposite ways, and both are visible here.

The cumulative k is higher than the band k at the same scale — at 10^6, 0.9489 against 0.9194. Including every smaller n blends in bands with larger k, and that raises the average. It is why the crossing is postponed from 62,000 to 196,000.

But the cumulative slope is steeper, 1.56 times so. Mixing bands of different scale adds dispersion, and added dispersion lowers a Weibull shape. The mean falls with M while the spread of means inside the sample stays roughly fixed in absolute terms, so the relative dispersion contributed by mixing grows, and k is pushed down harder than the band-wise drift alone would push it.

The misfit is constant. Only the test improves.

The last two columns of the table are the important ones, and they move in opposite directions.

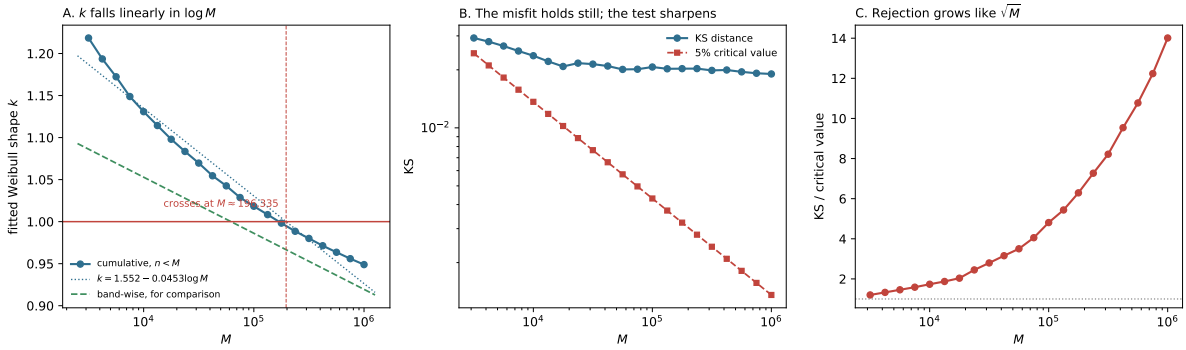
The KS distance barely changes: 0.0293 at $M = 3{,}162$, 0.0190 at $M = 10^6$. If anything it slightly improves. The critical value collapses, because it goes like $1/\sqrt{N}$: from 0.0244 down to 0.0014.

M	3,162	10^4	10^5	10^6
KS / critical value	1.2×	1.7×	4.8×	14.0×

So the Weibull is not getting worse as M grows. It is exactly as wrong at 10^6 as at 10^3, and the discrepancy was always there; what grows, like $\sqrt{M}$, is the power to see it. At

$M = 3{,}000$ the fit is marginal and a less demanding test would pass it. At $M = 10^6$ it fails by a factor of fourteen.

There is therefore no M at which the Weibull becomes correct, and no limiting k anywhere in this range. A shape that drifts is a shape that does not exist.



One caution on the formula. Linear-in-log M cannot hold indefinitely, since $k > 0$ always and the line reaches zero near $M \sim 10^{15}$. It is a description of the range measured, $10^{3.5}$ to 10^6, and not a law.

An aside, now that the periods reach 10^6

The mean period at $M = 10^6$ is 219.46, so mean$/\sqrt{M} = 0.2195$. Multiplying by $\log M$ gives 3.032, still rising; by $\sqrt{\log M}$ gives 0.8158, still falling. The growth of the mean remains bracketed strictly between $\sqrt{M}/\log M$ and $\sqrt{M}/\sqrt{\log M}$ at 10^6, as it was at 10^5. Another decade does not separate them.

8 Reproducing

```
python3 weibull_sweep.py     # periods, fits, KS, hazards for all seven bands
python3 weibull_plots.py     # the four band panels
python3 cumulative.py        # periods to 10^6, k(M) for 21 values of M
python3 cumulative_plot.py   # the three cumulative panels
tectonic weibull.tex
```

The Weibull maximum likelihood is written out directly rather than called from `scipy`, since the cumulative samples reach 10^6 points; it agrees with `scipy.stats.weibull_min.fit` to five decimal places on a test band ($k = 0.948014$ against 0.948032).

The periods are recomputed from scratch in every band. As a check, the band $[3\times10^5, 4\times10^5)$ reproduces 99,915 non-squares, mean 0.3343, shape 0.9480 and KS distances 0.0346 and 0.0210, matching the September figures to four decimal places.